Data Engineering
Assignment - 01

P. Vaishnavi
24N01A4431
CSD- 2nd year

1. How do Data Engineers and Data Scientists differ in their roles and tools used? Explain

→ Data engineers and Data Scientists play complementary but distinct roles in a data driven organization
→ While both work with data, they differ in focus, responsibilites & tools used.

Role Of a Data Engineer

A Data Engineer is responsible for designing, building and maintaining data infrastructure.
Their primary goal is to ensure that data is reliable scalable, secure and available for downstream use:

main responsibilites:-

- Collecting data from multiple sources
- Building and maintaining data pipelines
- Designing data lakes & data warehouses.
- Ensuring data quality, consistency.
- Monitoring and optimizing data system.

Common Tools used:-

- Programming: Python, Java, Scala, SQL
- Databases: MySQL, PostgreSQL, MongoDB
- Big data tools: Hadoop, Spark
- Data pipelines: Airflow, Kafka
- Cloud platform: AWS, Azure, Google Cloud.

Role Of a Data Scientist

A Data Scientist focuses on analyzing data to extract insights and build predictive models. Their work is closer to business and decision-making

main Responsibilites

- exploring and analyzing datasets
- Building statistical and ML models
- Performing hypothesis testing
- Creating data driven insights & predictions.
- Communicating results using visualizations and reports.

Common tools used:

Programming : Python, R, SQL
Libraries : Pandas, NumPy, Scikit-learn, Tensorflow
Visualization : Matplotlib, Seaborn, Tableau, Power BI
ML platforms : Jupyter Notebook, MLflow

Key Differences

	Data Engineer	Data Scientist
Focus	Data infrastructure and pipelines	Data analysis and modeling
Main Goal	Make data available and reliable	Extract insights & predictions
Type of Work	Backend and System oriented	Analytical and research-oriented
Output	Clean, structured datasets	Models, insights reports.

2. Data Engineering Life Cycle and Role of a Data Engineer.

→ The Data engineering life cycle describes the complete process through which data is generated, processed, stored and delivered for analysis and decision-making.

A Data engineering plays a critical role at every stage to ensure data is reliable, scalable and usable.

Data Generation

Data is created from various sources such as transactional systems, user interactions, sensors, logs and external APIs

Role of Data Engineer:-

Data engineers study data sources, understand data formats, volume, velocity, and quality and coordinate with source system owners to ensure data can be collected effectively.

Data Ingestion

In this stage, data is collected and transferred from source systems into storage platforms using batch or real-time pipelines.

Role of Data Engineer:

They design ingestion pipelines, select suitable ingestion methods, handle duplicate or late data and ensure fault-tolerant and reliable data tranfer

Data Storage

Ingested data is stored in databases, data lakes or data warehouses.

Role Of Data Engineer

They choose appropriate storage systems, design schemas, manage data partitioning and ensure scalability, durability and cost efficiency.

Data Transforming

Raw data is cleaned, validated, joined and aggregated to make it suitable for analysis

Role of Data Engineer

They implement transformation logic, maintain data quality rules, and ensure consistency while preserving raw data for future use

Data Serving :-

Processed data is made available to users through data warehouses, APIs, dashboards, or feature stores.

Role of Data Engineer :-

They optimize data models for querying, ensure fast data access and support analytical and application workloads.

Conclusion :-

The Data Engineering life cycle ensures smooth and reliable movement of data from source to consumption. At each stage, the data engineer acts as the architect and caretaker of data systems, enabling organizations to make accurate and informed decision.

3. Explain the key technical stakeholders of data engineering in detail both upstream and downstream How do data engineer acts as hub between data producers and data consumers?

Technical Stakeholders in Data Engineering and Role of Data Engineer

In data engineering, different technical stakeholders are involved in producing and consuming data. These stakeholders are divided into upstream stakeholders (data producers) and downstream stakeholders (data consumers). The data engineer acts as a central hub connecting both.

Upstream Stakeholders (Data Producers)

→ Application Developers

They build applications that generate user activity and transaction data.
Data engineers co-ordinate with them to understand data formats and schema changes.

→ Backend Engineers and DBAs

They manage databases, APIs and backend systems.
Data engineers ensure safe and efficient data extraction.

→ System and IoT Engineers

They produce machine data such as logs, metrics and sensor data.
Data engineers design scalable ingestion pipelines for this data.

Downstream Stakeholders (Data Consumers)

→ Data Analysts

They create reports and dashboards. Data engineers provide clean and structured datasets.

→ Data Scientists

They build analytical and machine learning models
Data engineers ensure reliable historical data & features.

→ ML Engineers and BI teams

They deploy models and support business decisions

Data Engineers ensure data accuracy, availability and performance.

Data Engineer as a Hub

The data engineer acts as a bridge by:

- Collecting data from upstream systems
- Cleaning, transforming and storing data.
- Delivering reliable data to downstream user.
- Maintaining data quality, security and governance.

Conclusion:

In modern data systems, the data engineer connects data producers and data consumers, ensuring smooth, reliable and trusted data flow across the organization.